

STAHLKRAFT

The cheapest way to shoot down drones.



**I build a cost-efficient, man-portable
robot that **detects**, **aims** and
shoots down drones.**

Summary

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The Problem

04

Drones are the #1 threat on the battlefield in Ukraine,
causing **70–80%** of all deaths.

A **\$500** drone can destroy a **\$10 million** tank.

No cheap way exists to detect and stop them.

Why Now

05

Jamming used to stop drones.

Now **nothing** can jam them.

Jamming works by cutting a drone's radio link — and the newest drones don't use one:



Wire-guided

Flown on a fiber, not radio.



Satellite-guided

Steered by Starlink, not radio.



Jam-resistant

Flies itself — no link at all.

Shooting them down is all that's left — but by hand, that **almost never works**.

The Solution

A robot that **detects**, **aims**
and **shoots down** the drone.

UNIT COST	~\$30,000
PER SHOT	~\$2
WEIGHT	Under 5 kg
WEAPON	Fits any rifle
DETECTION	Audio + camera + thermal, no radar
OPERATION	Day & night
TARGETS	Group 1–2 UAS *
FIRING	You approve each shot



* Group 1–2 — small & mid-size drones (FPV, quad, Mavic-class), not Shahed-class.

How It Works

07

01

Deployment

A soldier sets it up on a tripod — or it rides on a rover or a vehicle.



Tripod



Rover

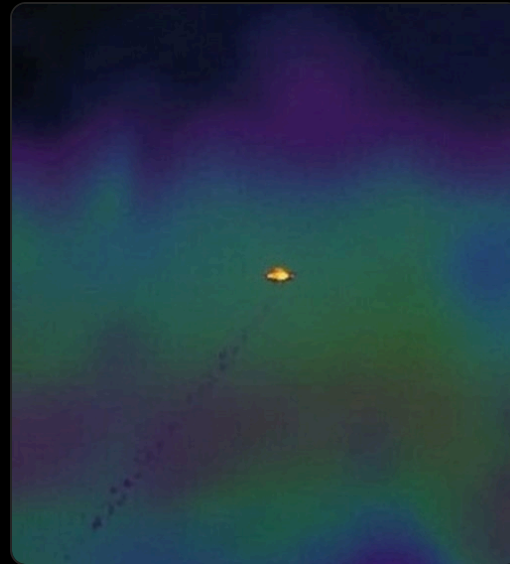


Vehicle

02

Detection

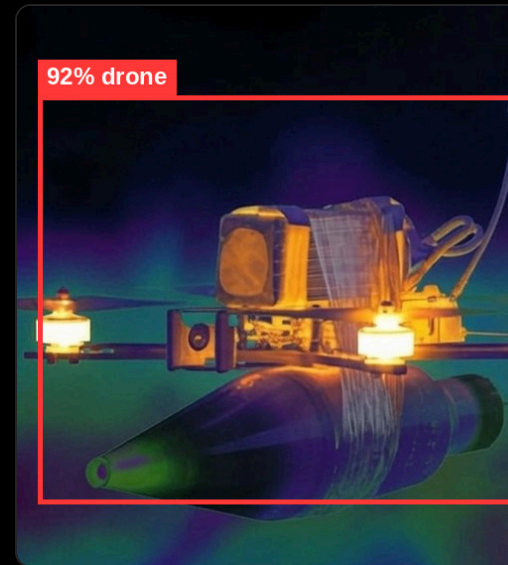
A **thermal lens** on a **step-and-stare ring** hunts for a drone's heat.



03

Confirmation & Tracking

The turret **locks on** to it and — as it gets closer — confirms if it's a **drone or a bird**.



04

Engagement

Approve or cancel from a remote or wrist device — for faster access.



Remote



Wrist device

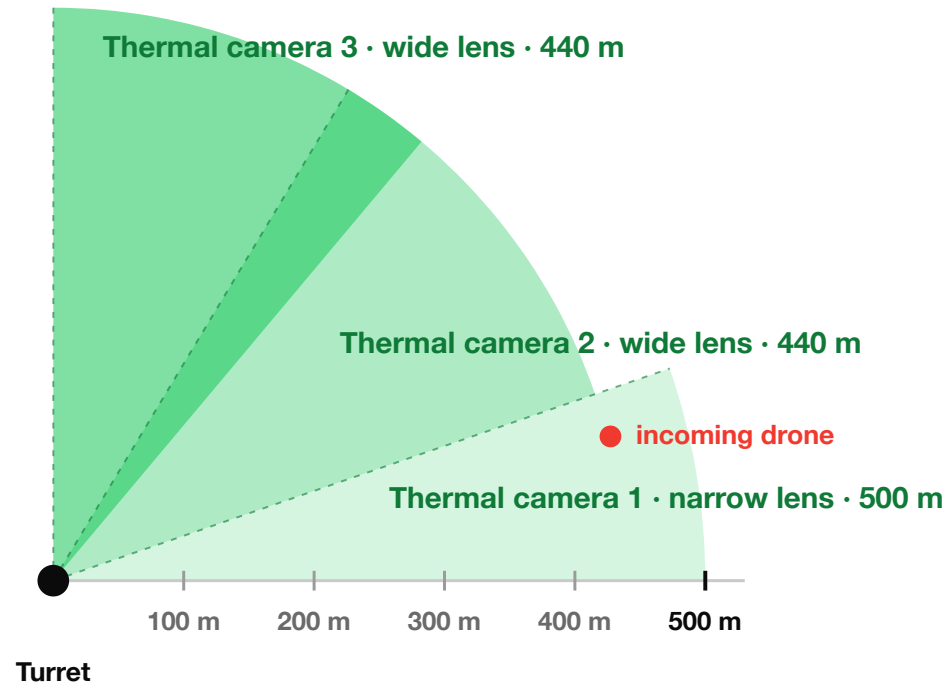
No radar. **90%** of radar performance at **10%** of the cost.

How It Works

08

No radar? Here's how.

Three rotating thermal cameras = a 360° heat dome.



1x

narrow thermal lens watches low — an FPV at 500 m. \$4K.

+2

wide thermal lenses cover up to straight overhead — no blind spot. \$8K.

~2 s

full sky re-scan — a drone is seen 4–5x before it arrives.

360°

Full heat dome for \$12K.

Prototype

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Link alternative: youtu.be/vkviBCPkYdU

It works.

It detects and tracks drones.

It fires. Tested with a real rifle.

Built by hand in one year, while in high school.

Not yet fired at a drone. Approval for live-fire testing is in process, but complicated by German regulation.

Competition

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Other companies have realised that shooting them down is **all that's left**.

Their approach in doing so, however, is **far too expensive** for small drones.

Let me explain:

Competition

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None of them detect small drones.

For detection, they rely on third-party \$100,000+ radars.



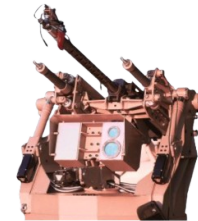
Tytan Technologies
Fixed-wing interceptor



Anduril
Reusable interceptor



P1-SUN
Interceptor drone



Sky Sentinel
Gun turret



SmartShooter Hopper
Remote weapon station

Without radar, none of the above work.

While the price and size of radar is justified against large threats like Shahed drones, it makes it impossible to resolve the small-drone threat at scale .

Competition

12

Without detection, even perfect aim is useless.

Smart scopes from **SmartShooter** and **ZeroMark** make every soldier a marksman — but only **after** the drone has been found. The math shows why that's too late:

THE PRODUCT



Fire-control smart scopes
SmartShooter SMASH · ZeroMark

STEP 1 · YOU HEAR IT

~150 m

distance at which a soldier
first hears an incoming FPV

STEP 2 · IT CLOSES FAST

55 m/s

attack speed of an FPV
(~200 km/h)

STEP 3 · THE RESULT

2–3 s

from first sound
to impact

Hear → locate in the sky → shoulder → aim → fire takes **5–10 s** — the drone always wins.

Aiming was never the bottleneck — detection is.

A smart scope only works when something else finds the drone first — and today that means a \$100,000+ radar.

Competition

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SYSTEM	REQUIRES RADAR	PRICE WITHOUT RADAR	PRICE WITH RADAR	COST PER ENGAGEMENT	DETECTION RANGE	MAN-PORTABLE
Stahlkraft turret	No	\$30K	N/A	\$2	500 m	Yes
Tytan METIS	Yes	~\$10–30K*	~\$150K*	~\$10–30K*	~1 km only with radar	Not man-portable with radar
Anduril Anvil	Yes	~\$50K*	~\$500K*	~\$50K*		
P1-SUN interceptor	Yes	~\$1–3K	~\$100K*	~\$1–3K		
Sky Sentinel	Yes	~\$150K	~\$300K*	~\$100*		
SmartShooter Hopper	Yes	~\$120K*	~\$300K*	~\$5*		

■ advantage ■ caveat ■ drawback · * estimate — Tytan, Anvil & Hopper prices aren't public.

We trade range for **scale**.

500 m still catches the drone in time — and only a \$30K, man-portable system is affordable enough to deploy at scale, **protecting troops everywhere they stand**.

One turret per **squad**.

Drone defence is the fastest-growing market in defence.

900K

soldiers in
Ukraine's army

=

90,000

squads of ten

×

\$30K

one turret each

=

\$2.7B

Ukraine alone

NATO Europe fields **2M+ troops** — and counter-drone budgets double to **\$24B** by 2030.

Every soldier on the front needs one — **no army has it yet**.

The Ask

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Raising **\$1M** to reach the first confirmed kill.



Stahlkraft becomes a US company.

Delaware C-Corp and I relocate — I turn **18 on 14 November 2026**, old enough to sign and incorporate.

USE OF FUNDS · 12 MONTHS

Team

\$450K

Two engineers and me, fully loaded. A working prototype is where solo ends.

Hardware

\$230K

Nine thermal cores, optics, compute and machining — three units, not one.

Live fire

\$200K

Range time, safety officers, ammunition — and the ~50 drones we destroy.

US setup

\$120K

Delaware entity, immigration, export counsel and weapons liability cover.

Hardware is under a quarter of it — the rest is people and range time.

\$1M

Why Me

16



Albert Ludemann, 17

Turning 18 on 14 November 2026.

Yale YYGS

summer program alumnus

Harvard CS50P

certified

Harvard AI50

certified

Self-taught

coding since primary school

Years of experience training image-classification AI models

Founded **jugendinvestiert.com** — a financial-education site in Germany.

A growing family of interceptors

Pictured: the threats — each one gets its own defense layer.

TODAY



Small & mid-size drones

→ our turret, working today

FUTURE



Shahed-class attack drones

→ our next interceptor

LONG-TERM



Cruise missiles

→ our missile interceptor

One different interceptor for each — covering all threats, from FPV drone to missile.

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